

Estimated time needed: 10 minutes

What are AI agent protocols?

AI agent protocols establish standards of communication among artificial intelligence agents and between AI agents and other systems. These protocols specify the syntax, structure and sequence of messages, along with communication conventions such as the roles agents take in conversations and what and how they respond to messages.

Agentic AI systems often can't talk. They're built by different providers using diverse AI agent frameworks and employing distinct agentic architectures. Real-world integration becomes a challenge, and coupling these fragmented systems requires tailored connectors for all possible types of agent interaction.

This is where protocols come in. They turn disparate multi-agent systems into an integrated ecosystem where AI-powered agents share a way of discussing, understanding and collaborating with each other.

Although agentic protocols are part of AI agent orchestration, they don't act as orchestrators. They standardize communication but don't manage agentic workflow coordination, execution and optimization.

Examples of AI agent protocols

Many protocols are still in their early stages, as they have yet to be widely used or applied on a large scale. This lack of maturity means that organizations must be prepared to act as early adopters, adjusting to breaking changes and evolving specifications.

As agentic technology evolves, new protocols might emerge. Here are some current common AI agent protocols:

- Agent Communication Protocol (ACP)
- Agent Network Protocol (ANP)
- Agent-Cloud Interactions (ACI) Protocol
- AgentAgent (AA) Protocol
- Model Context Protocol (MCP)
- AgentFramework Protocol (AFP)